

7.6 FUEL SYSTEM (see Fig. 7-8)

The fuel system comprises three subsystems, i.e. storage, supply, and indication. For re-fueling and defueling procedures, refer to Section 8.

7.6.1 Storage

Fuel is stored in underfloor compartments using three bladder-type fuel cells comprising two tanks; a supply tank and a main tank. The supply tank, located immediately aft of the landing gear forward cross tube, provides fuel for the engines. Two aft cells, located forward and aft of the landing gear aft cross tube, are interconnected with a large diameter tube to form the main tank. The main tank fuel is pumped to the supply tank with excess fuel returning via two overflow ports.

The fuel cells are normally vented overboard and are drained through separate valves accessible on the fuselage underside. The main tank fuel cells are refueled through a common filler neck located on the left side of the fuselage. The filler neck is designed to accept a standard gravity-fill type nozzle.

7.6.2 Supply

Fuel from the main tank is transferred to the supply tank through electric pumps. Fuel is then further pumped from the supply tank to each engine through separate supply systems, each comprised of an electrical supply tank pump and an electrical emergency fuel shutoff valve.

The fuel pump switches, labeled SUPPLY TANK - ENG 1, ENG 2 and MAIN TANK - 1, 2 are located on the overhead console. These switches power the fuel pumps from the Main Bus via separate circuit breakers (FUEL PUMPS - SUPPLY TANK, MAIN TANK) located on the center console switch panel. The EMERG FUEL VALVE - 1, 2 switches are also located on the center console switch panel and power the fuel valves from the Main Bus via separate circuit breakers (EMERG FUEL VALVE - 1, 2). The two EMERG FUEL VALVE switches are guarded to prevent accidental usage.

7.6.3 Indicators

Dual fuel quantity indicator (FUEL QTY) simultaneously indicates both the supply tank (S) and main tank (M) fuel quantities. Two fuel quantity transmitter probes are installed, one in each tank. The indicator is powered from the Main Bus via a fuse.

Dual fuel pressure indicator (FUEL PRESS) simultaneously indicates the left engine (1) supply pressure and the right engine (2) supply pressure in kp/cm^2 . A pressure transmitter is mounted in an adapter fitting installed immediately upstream of each engine-mounted fuel filter. The indicator is powered from the Main Bus via a fuse.

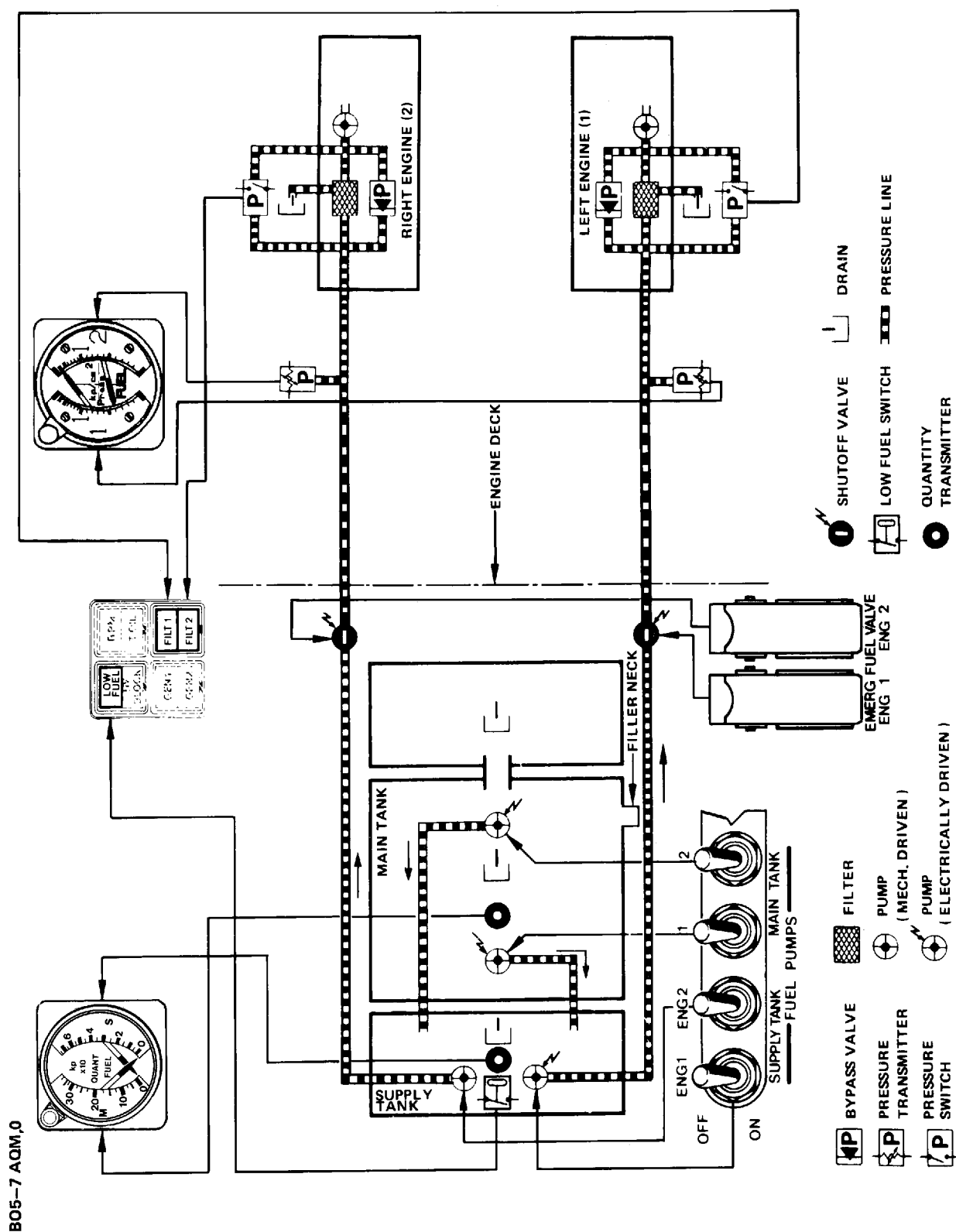


Fig. 7-8 Fuel System Schematic